

EV Society™ · EV.ENGINEER™

CubeTwin

The 12-week mission workbook

CubeSat energy, mission and reliability analysis

14 September – 6 December 2026

Mission / team: _____

Initial Draft · v0.01 · Scenario schema 1.0

Educational R&D prototype · Simulated data · Not flight software.

Last reviewed 12 September 2026

<https://aerospace.ev.engineer/space/cubesat>

PROJECT OVERVIEW

A spacecraft energy lab, in your browser.

CubeTwin is an educational web simulator that helps students model the electrical-energy behaviour of a CubeSat. It calculates how orbit sunlight, eclipse, solar generation, subsystem loads and battery characteristics affect State of Charge, voltage, temperature, mission activities and safe-mode decisions. The project is an EV Society™ initiative hosted on EV.ENGINEER™, with commercial engineering handled separately by iTelematics Software Private Limited.

A digital twin represents a specific physical system using traceable data. CubeTwin starts as a simulation platform; it is not connected to a spacecraft or calibrated against a physical test article.

HOW IT WORKS

How a CubeSat energy system works

The Electrical Power System (EPS) generates, stores and distributes the electricity that supports the mission. Follow its energy journey, then explore the equations.

01 / CUBESAT BASICS

A small spacecraft. A complete system.

A CubeSat is a small satellite built around standardized units and dispenser interfaces.

▼ Explanation, equation & example

Its payload performs the mission; its bus supplies power, computing, attitude control, communications and thermal support. A launch vehicle carries it to space.

3U ≈ three standard CubeSat units

TRY IT YOURSELF

Sketch a 3U mission. Identify the payload, EPS, OBC, ADCS and radio.

Continuously falling around Earth.

Altitude changes orbital period.

▼ Explanation, equation & example

With a mean Earth radius of 6,371 km, a circular orbit at 500 km takes about 94.5 minutes. Earth blocks direct sunlight during eclipse. This model starts each orbit in sunlight and uses a configurable eclipse at its end; real shadows depend on geometry and season.

$$T = 2\pi \sqrt{(R_E + h)^3 / \mu}$$

TRY IT YOURSELF

Predict the effect of moving from 400 to 600 km. Run both cases and compare periods.

Sunlight becomes usable power.

Solar cells convert light to electricity.

▼ Explanation, equation & example

Panel area, efficiency and incidence angle determine generation; PMAD conditions and distributes that power. Beginner mode takes peak power directly. Engineering mode exposes the same assumptions. Delivered solar power is zero during the simplified eclipse.

$$P_{\text{solar}} = G \times A \times \eta_{\text{cell}} \times \text{derating} \times \max(0, \cos \theta)$$

TRY IT YOURSELF

At 20 W peak and 90% PMAD efficiency, calculate the 18 W delivered to the bus.

Every activity has an energy cost.

Watts describe an instantaneous rate.

▼ Explanation, equation & example

Watt-hours describe energy accumulated over time. The nominal 8 W bus uses 4.67 Wh during a 35-minute eclipse before battery losses. Payload and downlink add 10 W and 6 W to the nominal bus. Essential-only operation uses 3 W.

$$P_{\text{net}} = P_{\text{solar, delivered}} - P_{\text{load}}$$

TRY IT YOURSELF

Calculate the battery withdrawal for 35 minutes at 8 W with 95% discharge efficiency: about 4.91 Wh.

Follow the energy in the battery.

SOC is $100 \times \text{stored energy} / \text{usable capacity}$; DoD is $100 - \text{SOC}$.

▼ Explanation, equation & example

Charging stores less energy than is supplied. Discharging removes more than the loads receive. The model clamps stored energy, records rejected charge at the upper limit and reports unmet demand when the battery is empty.

$$E_{\text{new}} = E + \eta_c P_{\text{net}} \Delta t / 3600 \text{ (charging)}; E + P_{\text{net}} \Delta t / (3600 \eta_d) \text{ (discharging)}$$

TRY IT YOURSELF

A 40 Wh battery starts at 80% SOC. Deliver 8 W for one hour at 95% efficiency: final energy ≈ 23.58 Wh and SOC $\approx 58.95\%$.

Voltage responds to load.

Open-circuit voltage comes from an editable illustrative pack lookup.

▼ Explanation, equation & example

Positive current means discharge. The current estimate uses requested battery power divided by OCV. Greater resistance produces more voltage sag. This diagnostic approximation does not determine SOC from voltage or solve a coupled circuit.

$$V_{\text{terminal}} = \text{OCV}(\text{SOC}) - I \times R_{\text{internal}}$$

TRY IT YOURSELF

At equal discharge current, double resistance and compare terminal voltage.

Heat is another state to watch.

The optional lumped model balances resistive heating, an assumed external heat input and heat exchange with a fixed bus temperature.

▼ Explanation, equation & example

Its parameters are educational assumptions. It excludes radiative geometry, cell chemistry and thermal runaway, and cannot assess battery safety.

$$\Delta T = \Delta t / C_{\text{th}} \times (I^2 R + Q_{\text{external}} - (T - T_{\text{bus}}) / R_{\text{th}})$$

TRY IT YOURSELF

Increase resistance while holding the other parameters constant. Compare the peak temperature.

Protect the reserve, then recover.

The deterministic controller defers noncritical payload/downlink below reserve and enters safe mode below its entry threshold.

▼ Explanation, equation & example

It exits only after the higher recovery threshold holds for the specified dwell time. This hysteresis prevents rapid switching. Completion, brownout intervals and per-orbit margins describe the simulated outcome.

$$\text{safe entry} < \text{reserve} \leq \text{recovery} \leq \text{maximum SOC}$$

TRY IT YOURSELF

Run Heater Stuck On. Find the first safe-mode event and explain the transition using sensed SOC.

Power-system background: [NASA Small Spacecraft Power Systems](#). Unit and interface guidance: [Cal Poly CubeSat Design Specification](#).

EDUCATIONAL MISSION SCENARIO

Meet your reference CubeSat.

Every value below is an illustrative simulator default, not a component recommendation or flight limit. The same values are editable in the simulation.

SPACECRAFT & ORBIT

A 3U spacecraft in Low Earth Orbit

A CubeSat uses standard units (U); a 3U model combines three units. Low Earth Orbit (LEO) is an orbit relatively close to Earth.

Altitude above Earth	500 km
Orbital period: one circuit	≈ 94.5 min
Sunlight / eclipse per orbit	≈ 59.5 / 35 min
Inclination: orbit-plane tilt	51.6°

POWER & BATTERY

A finite energy reserve

A watt (W) measures power; a watt-hour (Wh) measures energy. A 40 Wh energy capacity can ideally supply 8 W for 5 h before losses.

Peak solar / delivered power	20 / 18 W
Battery energy capacity	40 Wh
Initial State of Charge (SOC)	80%
Reserve / safe-mode entry	30% / 20%

MISSION PROFILE

Give the spacecraft a purpose

A mission profile schedules activities. The payload is the mission instrument; downlink sends its data to a ground station, an Earth-based radio facility.

Nominal / essential-only load	8 / 3 W
Payload activity: starts at 4 h	18 W · 10 min
Downlink activity: starts at 5 h	14 W · 8 min
Simulation duration / step	24 h / 10 s

12-WEEK STUDENT R&D ROADMAP

From first principles to an engineering prototype.

14 September – 6 December 2026. Expand each week to learn, calculate, build, verify and document. Progress is saved in this browser.

Your mission, one week at a time.

14 September – 6 December 2026 · Learn → Calculate → Build → Verify → Deliver
1 / 198 tasks complete · saved in this browser

01 Foundations & first prototype

Mission definition and CubeSat basics

14–20 September 2026

Learn

- ☒ spacecraft versus launch vehicle
- ☐ CubeSat units and the 3U reference mission
- ☐ payload, spacecraft bus and subsystems
- ☐ orbit, altitude, inclination and period
- ☐ why EPS is mission-critical.

Calculate

- ☐ Calculate the circular-orbit period at 500 km.

Build

- ☐ pure calculateOrbitalPeriod() function
- ☐ reference mission configuration
- ☐ first wireframe for desktop and 390 px mobile
- ☐ landing-page hero and project identity draft.

Verify

- ☐ dimensional/unit check
- ☐ hand result versus code within a defined tolerance
- ☐ invalid altitude inputs return controlled errors
- ☐ no existing /space content or routes regress.

Deliver

- ☐ Mission Definition v1
- ☐ assumptions register
- ☐ architecture sketch
- ☐ orbital-period unit tests
- ☐ initial website wireframe.

Power budget, sunlight and eclipse

21-27 September 2026

Learn

- ☐ watt, watt-hour, voltage, current and efficiency
- ☐ sunlight/eclipse energy cycle
- ☐ solar incidence angle and derating
- ☐ subsystem power budget
- ☐ difference between power and energy.

Calculate

- ☐ Calculate energy for a 35-minute, 8 W eclipse.

Build

- ☐ beginner eclipse-duration model
- ☐ solar peak-power model
- ☐ load scheduler
- ☐ net-power calculation
- ☐ preliminary sunlight/eclipsed orbit visual
- ☐ first SOC-versus-time line chart using a temporary ideal battery.

Verify

- ☐ solar power is zero during simplified eclipse
- ☐ net power equals delivered solar minus load
- ☐ energy-conservation test passes for a one-hour constant-load case
- ☐ chart labels and tooltips show units.

Deliver

- ☐ power budget table
- ☐ mission-mode table
- ☐ working solar/load prototype
- ☐ challenge problem/solution/novelty draft.

Battery MVP and ISAST submission

28 September–4 October 2026

Learn

- ☐ battery cell versus pack
- ☐ capacity, energy, SOC, DoD, charge/discharge efficiency
- ☐ why eclipse sizing matters
- ☐ why temperature and internal resistance matter
- ☐ simulator versus digital twin.

Calculate

- ☐ Integrate an 8 W load for one hour at 95% efficiency.

Build

- ☐ energy-integration battery model
- ☐ SOC limits, reserve warning and unmet-load accounting
- ☐ deterministic simulation loop
- ☐ Run/Pause/Reset controls
- ☐ a minimal demonstrable CubeTwin dashboard.

Verify

- ☐ SOC cannot exceed configured bounds
- ☐ charge/discharge losses have correct direction
- ☐ zero-load and zero-generation edge cases pass
- ☐ same scenario and seed produce identical output.

Deliver

- ☐ ISAST-ready PDF
- ☐ Battery Model v1
- ☐ demonstration video or screenshots
- ☐ submission receipt/reference recorded by the team.

The guide plans a concept submission milestone for 30 September. Confirm eligibility, deadline and submission requirements with the organizer; the full learning project continues to December.

Voltage, temperature and Stage Gate 1

5–11 October 2026

Learn

- ☐ OCV, terminal voltage and internal resistance
- ☐ voltage sag
- ☐ heat generation by I^2R
- ☐ model fidelity and assumptions
- ☐ verification versus validation.

Calculate

- ☐ Calculate voltage sag for 1 A at 0.12 Ω .

Build

- ☐ editable OCV/SOC curve
- ☐ terminal-voltage estimate
- ☐ optional lumped battery-temperature model
- ☐ warnings for undervoltage and temperature assumptions
- ☐ input help, glossary links and validation messages.

Verify

- ☐ increasing resistance increases voltage sag under equal load
- ☐ no divide-by-zero, NaN or infinite output
- ☐ thermal model remains stable at allowed time steps
- ☐ numerical functions have unit tests and explicit units.

Deliver

- ☐ Record your results, evidence, assumptions and one next improvement.

02 Mission behaviour & faults

Mission modes and energy-aware scheduling

12–18 October 2026

Learn

- ☐ spacecraft operations timeline
- ☐ payload/downlink scheduling
- ☐ energy margin
- ☐ safe mode and hysteresis.

Calculate

- ☐ Calculate the reserve-to-recovery energy gap.

Build

- ☐ mission state machine
- ☐ schedule editor or preset schedule cards
- ☐ payload and downlink activity markers on charts
- ☐ automatic low-SOC safe mode
- ☐ event log with reasons for every mode change.

Verify

- ☐ the state machine has no unreachable or accidental looping states
- ☐ safe mode reduces the configured load
- ☐ recovery requires the higher threshold and dwell time
- ☐ scheduled activities complete, defer or fail deterministically.

Deliver

- ☐ Record your results, evidence, assumptions and one next improvement.

Fault injection and diagnosis

19–25 October 2026

Learn

- ☐ fault, anomaly, failure and symptom
- ☐ truth state versus observed telemetry
- ☐ detection threshold and false alarm
- ☐ FDIR basics.

Calculate

- ☐ Predict the extra Wh from a 6 W heater for two hours.

Build

- ☐ at least six power fault scenarios
- ☐ truth/observed/detected separation
- ☐ annotated chart events
- ☐ fault comparison view: baseline versus faulty run
- ☐ simple rule-based detection, not ML.

Verify

- ☐ each injected fault begins and ends at the configured time
- ☐ baseline results do not change when no fault is enabled
- ☐ detectors are tested for expected true/false outcomes
- ☐ imported scenarios cannot inject code or unsafe markup.

Deliver

- ☐ Record your results, evidence, assumptions and one next improvement.

Telemetry and reliability metrics

26 October–1 November 2026

Learn

- ☐ telemetry, sampling interval and time series
- ☐ energy throughput and equivalent full cycles
- ☐ mission success criteria
- ☐ margin and uncertainty.

Calculate

- ☐ Calculate equivalent cycles from charge/discharge throughput.

Build

- ☐ telemetry table and charts for SOC, power, voltage and temperature
- ☐ metrics summary
- ☐ activity completion score
- ☐ JSON/CSV export with units and metadata
- ☐ accessible chart summaries for screen readers.

Verify

- ☐ sums/integrals match raw telemetry
- ☐ exported row count and timestamps are correct
- ☐ units appear in headers
- ☐ large 24-hour runs stay within mobile memory limits.

Deliver

- ☐ Record your results, evidence, assumptions and one next improvement.

3D mission experience and Stage Gate 2

2–8 November 2026

Learn

- ☐ coordinate frame as a reference for position/orientation
- ☐ visual scale versus physical scale
- ☐ why 3D visuals must not be treated as analysis results
- ☐ accessible alternatives to WebGL.

Calculate

- ☐ Convert one sample time into orbital phase.

Build

- ☐ optimized 3D Earth, orbit path, moving CubeSat and day/night indication
- ☐ synchronized simulation clock and 3D state
- ☐ camera reset and pause
- ☐ quality setting or automatic mobile simplification
- ☐ static 2D fallback when WebGL is unavailable
- ☐ reduced-motion mode.

Verify

- ☐ the 3D scene agrees with sunlight/eclipsed state from the numerical engine
- ☐ no duplicated physics is implemented inside the renderer
- ☐ touch controls do not block page scrolling
- ☐ 360, 390, 412 and 768 px layouts remain usable
- ☐ keyboard and screen-reader users can obtain the same numerical information.

Deliver

- ☐ Record your results, evidence, assumptions and one next improvement.

03 Uncertainty & release

Monte Carlo uncertainty

9–15 November 2026

Learn

- ☐ uncertainty range versus probability distribution
- ☐ random seed and reproducibility
- ☐ deterministic run versus Monte Carlo run
- ☐ confidence interval and failure probability.

Calculate

- ☐ Compute a completion fraction and compare its confidence interval.

Build

- ☐ uncertainty inputs for capacity, load, solar output and resistance
- ☐ seeded batch runner in a worker
- ☐ histogram or distribution summary
- ☐ probability of falling below reserve and completing activities.

Verify

- ☐ fixed inputs collapse to the deterministic result
- ☐ same seed reproduces the same distribution
- ☐ sample count is capped for mobile safety
- ☐ claims state simulated probability, not real mission probability.

Deliver

- ☐ Record your results, evidence, assumptions and one next improvement.

Model validation and learning workbook

16–22 November 2026

Learn

- ☐ requirement, verification and validation
- ☐ test case, expected result and tolerance
- ☐ reference data and calibration
- ☐ model limitation.

Calculate

- ☐ Close the generated, consumed and stored energy balance.

Build

- ☐ validation cases with hand calculations
- ☐ requirement-to-test traceability table
- ☐ tutorial modules, exercises and answer guidance
- ☐ print stylesheet and printable workbook cover
- ☐ references and last-reviewed date.

Verify

- ☐ orbit-period result is compared to the analytical formula
- ☐ constant-power battery cases match analytical energy change
- ☐ energy accounting reports numerical residual
- ☐ all external sources are authoritative and linked
- ☐ print preview has no clipped tables, invisible text or unnecessary dark backgrounds.

Deliver

- ☐ Record your results, evidence, assumptions and one next improvement.

SEO, AI search and production quality

23–29 November 2026

Learn

- ☐ Review model assumptions and the weekly learning material.

Calculate

- ☐ Check canonical URLs and a public crawler response.

Build

- ☐ canonical metadata for /space/cubesat
- ☐ sitemap and robots policy
- ☐ Open Graph/Twitter image at 1200×630
- ☐ semantic headings and internal links
- ☐ JSON-LD entity graph
- ☐ visible FAQ with concise answers
- ☐ optional /llms.txt summary as a supplemental discovery aid
- ☐ analytics events for simulator start, scenario completion, workbook print and outbound ecosystem links, without personal data.

Verify

- ☐ no noindex or accidental crawler block
- ☐ canonical and Open Graph URLs use the exact production path
- ☐ structured data matches visible content and has no fabricated ratings, partners or awards
- ☐ page is server-rendered enough for crawlers to understand without running the simulator
- ☐ broken-link, metadata and sitemap tests pass
- ☐ Search Console, Bing tools and schema validators are documented as post-deployment checks.

Deliver

- ☐ Record your results, evidence, assumptions and one next improvement.

Release, demonstration and handover

30 November–6 December 2026

Learn

- ☐ Review model assumptions and the weekly learning material.

Calculate

- ☐ Compare final demonstration results against requirements.

Build

- ☐ polish the default, low-energy and fault scenarios
- ☐ complete README, model assumptions, validation report and contributor guide
- ☐ add version number and changelog
- ☐ create a 3–5 minute beginner demonstration script
- ☐ complete technical and mentor review
- ☐ deploy through the repository's existing verified process.

Verify

- ☐ full tests, build, lint/type-check and mobile end-to-end tests pass
- ☐ Lighthouse lab targets are met or documented
- ☐ no console errors, hydration warnings or broken downloads
- ☐ simulator output is deterministic when Monte Carlo is off
- ☐ disclaimers and organization roles are accurate
- ☐ rollback procedure and known limitations are documented.

Deliver

- ☐ Record your results, evidence, assumptions and one next improvement.

MISSION & RELIABILITY ANALYSIS

Test the mission when conditions change.

Reliability means performing the required functions under stated conditions. Compare a fault experiment with the same mission without faults, then explore input uncertainty.

BASELINE VS. EXPERIMENT

Change one thing. See the consequence.

The baseline uses the current mission configuration with all faults removed. Both runs use identical models and schedules.

No faults injected · baseline active

Whole-run metric	Baseline	Experiment
Minimum SOC	80.00 %	80.00 %
Final SOC	91.81 %	91.81 %
Energy margin	79.99 Wh	79.99 Wh
Below reserve	0.00 min	0.00 min
Minimum voltage	7.92 V	7.92 V
Safe-mode entries	0	0
Unmet energy	0.00 Wh	0.00 Wh

Fault injection is not proof of detection. The SOC discrepancy detector uses simulated truth; real detection requires independent measurements.

Inspect experiment time0.00 h



GO BEYOND A SINGLE RUN

How sensitive is your mission?

Sample battery capacity, solar output, total subsystem load and resistance independently from uniform $\pm$ ranges. Seeded trials run in a worker. Bounds are clipped to the scenario limits; temperature and faults stay fixed.

A trial succeeds when all activities complete with no unmet load or undervoltage. This is a simulated probability under stated assumptions.

▼ Verification, model assumptions & limitations

Check the evidence

Verification checks that code implements its equations correctly. Validation checks whether those equations represent the intended physical system.

Compare orbital period and constant-power battery cases with hand calculations.

Close the energy balance: initial stored + generated – served load – rejected energy – losses – capacity-fault removal – final stored ≈ 0 .

Inspect event timestamps, activity outcomes and changes at different time steps.

Reproduce the exported scenario with the same random seed.

Simulation results are not flight evidence.

No physical test article or measured mission data calibrates this prototype. It cannot demonstrate flight readiness, qualification or battery safety.

Model assumptions

Circular two-body orbit: Earth mean radius 6371 km; gravitational parameter 398600.4418 km³/s². Inclination is descriptive in duration mode.

Each orbit begins in sunlight; eclipse is a configurable interval at the end. No seasonal geometry, attitude dynamics or perturbations.

Battery energy in Wh; power in W; time in seconds. Positive battery power/current means discharge. Efficiency losses are included.

OCV is an illustrative editable pack lookup. Current = requested battery power / OCV; voltage sag and I^2R temperature are diagnostic approximations, not a coupled electrical solver.

Constant capacity except timed fault windows. Capacity loss removes excess stored energy explicitly; recovery never creates energy.

SOC bias affects onboard decisions. Telemetry dropout affects ground observations only. Bias detection uses truth as a teaching oracle, unavailable to real operators.

A deferred activity is skipped, not rescheduled. Mission completion requires every activity completed, no unmet load and no undervoltage interval.

Lumped thermal model is uncalibrated; no radiation, electrochemistry, battery safety prediction, SOH or RUL estimate.

Record your prediction. Explain the result.

Print the same lessons and project checklist, save a scenario or keep local experiment notes. Use the browser print dialog to save a PDF.

THE CUBETWIN FIELD NOTES

Build understanding. Keep the evidence.

A printable companion to the same lessons, equations and 12-week project. Capture your predictions before you run, then explain what changed.

Eight experiments to make it yours

Calculate the period of the 500 km reference orbit and repeat at 600 km.

Calculate battery energy used during a 35-minute eclipse at 8 W, including 95% discharge efficiency.

Compare the baseline with a 35% solar degradation fault. Explain the change in minimum SOC.

Move the downlink activity into a sunlight interval. Record whether the activity completes.

Trigger safe mode. Record entry, recovery threshold, dwell time and exit.

Compare true SOC and observed SOC with a +15 percentage-point sensor bias.

Run 50 Monte Carlo trials twice with the same seed. Explain reproducibility and uncertainty.

Write a conclusion: assumptions, analytical verification, limitations and next experiment.

Your prediction, observations & engineering conclusion

QA: increased resistance produces greater voltage sag.

Big ideas. Plain language.

A searchable field guide to spacecraft, energy and simulation.

Spacecraft

A vehicle or engineered system designed to operate in space. A satellite is one type of spacecraft.

Launch vehicle

The rocket that carries a spacecraft from Earth toward its deployment orbit.

Satellite

An object orbiting another body. In this guide it means an artificial spacecraft orbiting Earth.

CubeSat

A small satellite built around standardized unit sizes and dispenser interfaces.

1U

One CubeSat unit, approximately a 10 cm cube under the current CubeSat design specification; larger formats combine units.

3U CubeSat

A CubeSat roughly three units long, providing more volume than a 1U spacecraft.

Spacecraft bus

The supporting platform—power, computing, communications, structure, thermal and attitude control—that enables the payload.

Payload

The instrument or experiment that performs the mission's main purpose.

Mission

The complete objective and lifecycle of the spacecraft, including design, testing, launch and operations.

LEO (Low Earth Orbit)

An orbit relatively close to Earth, commonly used by CubeSats.

Altitude

Height above a reference Earth surface.

Orbit

The curved path followed by a spacecraft under gravity.

Orbital period

Time required to complete one orbit.

Inclination

Angle between the orbital plane and Earth's equatorial plane.

Orbit plane

The geometric plane containing the spacecraft's idealized orbit.

Ephemeris

Time-tagged information describing where a celestial body or spacecraft is expected to be.

Perturbation

A force/effect that makes a real orbit differ from an ideal two-body orbit.

Eclipse

Interval when Earth or another body blocks direct sunlight from the spacecraft.

Sun vector

Direction from the spacecraft/reference frame toward the Sun.

Attitude

Orientation of a spacecraft in space; it is different from altitude.

Reference frame

Coordinate system used to describe position, velocity or attitude.

Ground station

Earth-based equipment that communicates with the spacecraft.

Pass

Period when the spacecraft is visible to a ground station and communication may be possible.

Telemetry

Measurements and status data transmitted from the spacecraft.

Telecommand

Instruction transmitted from the ground to the spacecraft.

EPS (Electrical Power System)

Subsystem responsible for electrical generation, storage, control and distribution.

PMAD

Power Management and Distribution; circuits that condition, control and deliver electrical power.

OBC

On-Board Computer; runs flight software and coordinates spacecraft functions.

C&DH

Command and Data Handling; receives commands, handles data and supports onboard processing/storage.

ADCS

Attitude Determination and Control System; estimates and controls orientation.

TT&C

Telemetry, Tracking and Command; communication functions used to monitor and operate the spacecraft.

Thermal control

Methods used to keep spacecraft components within required temperature ranges.

Reaction wheel

Motor-driven wheel that changes spacecraft attitude through angular momentum exchange.

Magnetorquer

Electromagnetic actuator that interacts with Earth's magnetic field to create control torque.

Safe mode

Predefined low-risk state that protects essential functions after a serious anomaly or low-energy condition.

Redundancy

Extra hardware, software or paths intended to tolerate a failure.

FDIR

Fault Detection, Isolation and Recovery; identify an abnormal condition, locate likely cause and perform/advise a bounded response.

Voltage (V)

Electrical potential difference that helps drive current.

Current (A)

Rate of electric-charge flow.

Power (W)

Instantaneous rate of energy use or generation; $P=VI$ in a simple DC case.

Energy (Wh)

Power accumulated over time; 10 W for 2 hours is 20 Wh.

Power budget

Table comparing expected generation and consumption in each mode.

Energy budget

Calculation of generated and consumed energy over a period such as one orbit.

Solar cell

Semiconductor device converting light into electricity.

Solar array

Connected solar cells/panels that generate spacecraft power.

Incidence angle

Angle between incoming sunlight and the solar-panel normal.

Derating

Intentionally reducing ideal performance to account for real losses, aging, temperature and uncertainty.

BOL

Beginning of Life; performance near the start of the mission.

EOL

End of Life; expected performance near mission end after degradation.

Battery cell

Smallest electrochemical unit in a battery.

Battery pack

Connected cells plus protection, structure, sensing and often management electronics.

Li-ion

Lithium-ion rechargeable battery chemistry family widely used where high energy density is valuable.

BMS

Battery Management System; electronics/software that monitor and protect a battery within its design limits.

Capacity (Ah)

Amount of electric charge a battery can deliver under specified conditions.

Energy capacity (Wh)

Approximate stored electrical energy; depends on voltage and charge capacity.

SOC (State of Charge)

Estimated percentage of presently available battery charge/energy.

DoD (Depth of Discharge)

Percentage of capacity removed; approximately 100% minus SOC under the simplified definition.

SOH (State of Health)

Estimate of present battery capability relative to a defined new/reference condition.

RUL (Remaining Useful Life)

Estimate of time or cycles before a defined end-of-use criterion; highly dependent on data and assumptions.

OCV (Open-Circuit Voltage)

Battery voltage measured/estimated without load after suitable rest conditions.

Internal resistance

Effective opposition inside the battery that creates voltage drop and heat under current.

Voltage sag

Temporary terminal-voltage reduction when load current rises.

Charge efficiency

Fraction of charging energy stored by the battery in the model.

Discharge efficiency

Fraction accounting for energy lost while delivering power.

C-rate

Charge/discharge current normalized to battery capacity; 1C nominally corresponds to using rated capacity in about one hour under stated conditions.

Energy throughput

Total charged and discharged energy accumulated over time.

Equivalent Full Cycle (EFC)

Partial cycles combined into the energy equivalent of one full charge-discharge cycle.

Capacity fade

Reduction in available battery capacity with aging/use.

Thermal runaway

Dangerous self-heating condition in which heat-generating reactions accelerate; CubeTwin does not predict or certify this phenomenon.

Model

Mathematical/software representation of selected real-world behaviour.

Simulation

Running a model over time under defined inputs and assumptions.

Digital twin

A digital representation connected to the state/history of a particular physical system. CubeTwin begins as a physics-based digital-twin simulation; it becomes a calibrated twin only when linked to traceable data from a specific test article or spacecraft.

Time step

Simulated time between numerical updates.

State variable

Quantity carried from one time step to the next, such as SOC or temperature.

Boundary condition

Defined environmental or interface condition used by a model.

Scenario

Complete set of mission inputs, schedule, faults and duration.

Fault

Defined defect or abnormal condition that may cause an undesired effect.

Anomaly

Observed behaviour that differs from expectation; it does not automatically prove the root cause.

Failure

Inability to perform a required function within specified limits.

Fault injection

Deliberately adding a simulated fault to test detection and response.

Ground truth

Hidden, authoritative simulated physical state used to assess an observer/detector.

Threshold

Value that triggers a status, warning or action.

Hysteresis

Different entry and exit thresholds used to prevent rapid toggling.

Energy margin

Energy remaining beyond what is required for the planned activity/reserve.

Brownout

Voltage/power condition too low for correct operation of one or more loads.

Reliability

Probability or ability to perform required functions for a stated time under stated conditions.

Monte Carlo simulation

Repeated runs with sampled uncertain inputs to explore a distribution of outcomes.

Seed

Starting value that makes pseudo-random simulations reproducible.

Verification

Evidence that the software/model was implemented according to its stated requirements—"built correctly."

Validation

Evidence that the model is appropriate for the intended use—"the right model for this purpose."

Calibration

Adjusting model parameters using traceable observed/test data.

Uncertainty

Quantified lack of exact knowledge about inputs or model behaviour.

Tolerance

Allowed difference between expected and calculated results.

Digital thread

Traceable connection among requirements, parameters, models, tests, data and decisions.

HIL (Hardware-in-the-Loop)

Test setup where real hardware interacts with a simulated environment/system.

TRL (Technology Readiness Level)

Scale used to describe technology maturity; do not assign a TRL without criteria and evidence.

HUMS

Health and Usage Monitoring System; observes condition and usage to support diagnostics, prognostics and maintenance.

BEGINNER QUESTIONS

Answers before your next experiment.

Key definitions and model boundaries, explained plainly.

▼ What is CubeTwin?

CubeTwin is an educational CubeSat electrical-energy simulator and 12-week learning portal. It uses transparent models and simulated data to connect orbit, sunlight, battery behaviour, mission activities and power faults.

▼ How does a CubeSat battery survive eclipse?

During sunlight, the solar array supplies spacecraft loads and charges the battery. In eclipse, the battery supplies those loads. Teams size the energy store, account for conversion losses, protect a reserve and schedule demanding activities around available energy.

▼ What is a CubeSat electrical power system?

The Electrical Power System (EPS) generates, stores, manages and distributes electrical power. It includes solar generation, batteries and power management and distribution (PMAD) electronics.

▼ Do I need aerospace or coding experience?

No. Start with the bundled 3U LEO Beginner Mission, run the simulation and follow the plain-language lessons. The 12-week guide introduces the science, hand calculations, software and verification in stages.

▼ Is CubeTwin connected to a real spacecraft?

No. All data is simulated in your browser. CubeTwin is not flight software, live telemetry, certified analysis or a validated operational digital twin. Calibration against a specific physical test article is future work.

▼ What does the Monte Carlo probability mean?

It is the fraction of seeded simulated trials completing all activities without unmet load or undervoltage under the selected uniform input uncertainty. The confidence interval describes sampling uncertainty. Neither value is a prediction of real spacecraft reliability.

▼ Can I save my work and use the workbook offline?

Export the complete scenario or run as JSON, export telemetry as CSV, or use Print the Workbook and choose Save as PDF. Workbook notes and weekly progress are stored only in this browser when local storage is available. Downloads can be used offline; the portal is not an offline application.

▼ Who is responsible for CubeTwin?

CubeTwin is an EV Society™ education and research initiative hosted on EV.ENGINEER™. Commercial engineering is handled separately by iTelematics Software Private Limited under explicit agreements. UFlight™ is referenced for the broader aerospace HUMS ecosystem.

REFERENCES & LEARNING RESOURCES

Go straight to the source.

Authoritative starting points for your reading and future model comparisons. Record the document revision used in your report.

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|----|--|---|
| 01 | NASA · CubeSat 101
Mission development and beginner systems engineering |  https://www.nasa.gov/wp-content/uploads/2017/03/nasa_csli_cubesat_101_508.pdf |
| 02 | Cal Poly · CubeSat Design Specification
CubeSat units and preliminary interface requirements |  https://www.cubesat.org/cubesatinfo |
| 03 | NASA · Small Spacecraft Power Systems
Electrical power generation, storage and distribution |  https://www.nasa.gov/smallsat-institute/sst-soa/power-subsystems/ |
| 04 | ECSS · Li-ion Battery Testing Handbook
Battery test considerations; consult applicable revisions |  https://ecss.nl/wp-content/uploads/2016/06/ECSS-E-HB-20-02A1October2015.pdf |

05 **NASA · General Mission Analysis Tool**

<https://gmat.atlassian.net/wiki/spaces/GW/overview>

A future reference for higher-fidelity orbital comparisons

06 **Orekit · Official documentation**

<https://www.orekit.org/>

A future reference for orbit and shadow modelling

Content last reviewed: 12 September 2026 · Initial Draft v0.01 · Scenario schema 1.0. Applicable launch-provider requirements supersede preliminary interface guidance.

PROJECT OWNERSHIP & RESPONSIBILITIES

Clear roles. An educational purpose.

CubeTwin is an EV Society™ education and research initiative hosted on EV.ENGINEER™.

EDUCATION & RESEARCH

EV Society™

The education and research initiative behind CubeTwin.

EV Society [↗](#)

EV.ENGINEER >

COMMERCIAL ENQUIRIES

iTelematics

Commercial engineering products, implementation services and customer engagements, where applicable, are handled separately by iTelematics Software Private Limited under explicit agreements.

iTelematics [↗](#)

RELATED ECOSYSTEM

UFlight™

UFlight™ is referenced within the broader ecosystem for Health and Usage Monitoring Systems related to aerospace and autonomous platforms.

UFlight [↗](#)

🛡 Educational R&D prototype · Simulated data · Not flight software.

These references do not imply a partnership, endorsement, certification, ISRO affiliation or government approval.

